

Lab 9: BB84 on the Bench

Quantum key distribution (QKD) on a tabletop kit

1 ECTS · in person at i-LAMP

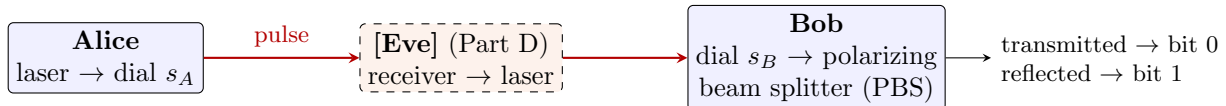
Prerequisites: *Fundamental Concepts of Quantum Technologies* and *Quantum Communication* (lecture series) — nothing else.

What you need: the Thorlabs EDU-QCRY1 kit with Alice, Bob and Eve modules (set up by the staff); Alice’s black 8-sided die, red 6-sided dice for Bob and Eve; the worksheet (pages 4–5); a pen.

Safety: class 2 laser, 635 nm, pulses of at most 1 mW. Never stare into the beam or its reflections; if dazzled, close your eyes and turn away (manual Ch. 13).

Source: Thorlabs, *EDU-QCRY1 Quantum Cryptography Demonstration Kit User Manual*, Rev D (2026), “manual §x”.

Goal: run BB84, the QKD protocol of the *Quantum Communication* course, on an optical bench, and find out what the bench can and cannot show.



Part A — Does the channel work?

Intuition. Alice’s laser emits horizontally polarized pulses. In Dirac notation (from *Fundamental Concepts*) we call horizontal polarization $|H\rangle$ and vertical $|V\rangle$, and use them as the bits: $|H\rangle \leftrightarrow 0$, $|V\rangle \leftrightarrow 1$. Light polarized at angle θ (from the horizontal, counter-clockwise as seen from behind Alice’s laser, looking along the beam) is $|\theta\rangle = \cos\theta|H\rangle + \sin\theta|V\rangle$. A **basis** here is a pair of perpendicular polarizations: the $+$ basis $\{0^\circ, 90^\circ\}$ and the $\times$ basis $\{-45^\circ, 45^\circ\}$.

The half-wave-plate rule. A half-wave plate mirrors the polarization direction about the plate’s axis; each module’s dial reads the result directly in polarization degrees (the dial shows twice the axis angle). *Alice’s dial turns horizontal light to the dial angle:* her axis sits at $s_A/2$ and mirrors 0° to s_A , so her pulse leaves as $|s_A\rangle$. *Bob’s dial faces the other way* (away from Alice, manual §7.4), so it reads with the opposite sign: his axis sits at $-s_B/2$ and mirrors $|s_A\rangle$ to $|-(s_B - s_A)\rangle = |-(s_A + s_B)\rangle$, and that is the light reaching his PBS.

Formula. The polarizing beam splitter (PBS) transmits $|H\rangle$ to one sensor (bit 0) and reflects $|V\rangle$ to the other (bit 1). The Born rule (*Fundamental Concepts*) gives $P(\text{transmitted}) = |\langle H|\theta'\rangle|^2 = \cos^2\theta'$ for light at angle θ' ; for a bright beam this is Malus’s law, the transmitted fraction of the intensity. With $\theta' = -(s_A + s_B)$:

$$P(\text{transmitted}) = \cos^2(s_A + s_B), \quad P(\text{reflected}) = \sin^2(s_A + s_B).$$

Bob measures in two steps: his plate maps the basis he chose onto the H/V split, then the PBS splits it and the sensor absorbs the light. **Number:** $s_A = 45^\circ$, $s_B = 45^\circ$ gives $\cos^2 90^\circ = 0$: reflected, bit 1.

Predict → measure → compare (worksheet W1)

- Predict** all 8 cases, $s_A \in \{-45^\circ, 0^\circ, 45^\circ, 90^\circ\}$ and $s_B \in \{0^\circ, 45^\circ\}$: transmitted (T), reflected (R), or *both* where the formula gives $\frac{1}{2}$.
- Measure** in *adjustment mode* (yellow side LED: when both sensors receive equal light, both blue LEDs light; manual §7.2.3). Check the alignment with the fine-tuning steps of manual §7.4, then fire one pulse per case.
- Compare.** All 8 cases must match before Part C. If one fails, call the staff (troubleshooting: manual Ch. 11).

The bench is not the model: misalignment

Our formula assumes perfect plates, dials and sensors, with no loss and no noise. A plate or sensor slightly off moves a “certain” case away from exactly 0 or 1, so some pulses light the wrong LED. In Part C this shows up as *errors without any Eve*. That is why all 8 cases must pass first, and why nobody leans on or moves the table during a run (manual §7.4).

Part B — Who flips the coin?

In 4 of the 8 cases Alice and Bob use different bases, the formula gives $\frac{1}{2}$, and in adjustment mode *both* LEDs light. **Predict:** a real photon would give a single random bit, 0 or 1 with probability $\frac{1}{2}$ each. **Measure:** keep one $\frac{1}{2}$ case, switch to *measuring mode* (green side LED), fire several pulses and tally T and R in W1. **Compare** with what adjustment mode showed for the same settings.

Counterintuitive — and Real

The pulse is classical light: at a $\frac{1}{2}$ case it really *does* split, half the intensity to each sensor — that is what adjustment mode shows. In measuring mode the kit's electronics **picks one LED at random** (manual §7.2.3). A single photon would not split: it reaches one sensor or the other, and nothing on the bench decides in advance which. The statistics are the same Born numbers, but here the coin is electronic. **The kit is an analogy of BB84**, not a quantum channel.

Part C — Can Alice and Bob agree on a secret?

Intuition. When Alice's and Bob's bases match, Bob's bit equals Alice's (Part A); when they differ, Bob's bit is a coin flip. So they keep only the matched pulses, and they can find those by talking in public without ever saying a bit.

1. **Dice, before firing.** For each of $N = 52$ pulses, Alice rolls the 8-sided die (each face shows a basis and a bit) and Bob the 6-sided die (each face shows a basis) (manual Ch. 10). Fill rows 1–3 of W2. Use the dice, not your head: people are poor random number generators.
2. **Transmit** in measuring mode. Alice sets $+$: 0° for 0, 90° for 1; $\times$: -45° for 0, 45° for 1. Bob sets 0° for $+$ and 45° for $\times$, and records transmitted = 0, reflected = 1.
3. **Sift.** *Sifting* means keeping only the pulses where Alice's and Bob's bases agree. They read out their bases in public (never their bits) and tick “same basis”. The bases match with probability $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$, so the sifted length S is about $N/2 = 26$, and $P(S \geq 20) = 96.5\%$.
4. **Test.** Publicly compare $t = 6$ sifted bits at random positions, chosen after the transmission; mark each one = (same bit) or $\neq$ (different bit) in the Test row, and drop them from the key. The *QBER* (quantum bit error rate) is the probability that Bob's sifted bit differs from Alice's; its estimate is (number of errors)/ t . Predicted without Eve on an aligned bench: QBER = 0.
5. **Encrypt — only if the test shows 0 errors.** The *one-time pad* adds each message bit m to a key bit k modulo 2 (XOR: $0 \oplus 0 = 1 \oplus 1 = 0$, $0 \oplus 1 = 1 \oplus 0 = 1$), $y = m \oplus k$, with a key as long as the message, random, and used only once. Since $k \oplus k = 0$, Bob recovers $(m \oplus k) \oplus k = m$. Code a 4-letter word with the 5-bit alphabet (manual Ch. 9; W5): 20 key bits from the untested sifted bits. The ciphertext may travel in public (manual §8.2); Bob decrypts it.

Budget: 20 key bits + 6 test bits need $S \geq 26$, which 52 pulses give with probability 55.5%. If you fall short, fire more pulses (manual §8.1; W2 has room for pulses 53–70); never reuse key bits. *This is the BB84 order: sift, test, then encrypt. Manual Ch. 8 encrypts first, as a first-pass teaching sequence (manual Ch. 10).*

The $\times$ labels

In the $\times$ basis, bit 0 is the “ -45° ” setting on Alice's dial and bit 1 the “ 45° ” setting; Bob's “ 45° ” setting reads transmitted as 0. Which diagonal is called “0” is a naming convention with no physical effect: renaming both would change no LED and no error count.

Part D — Can Eve hide?

Intuition. Eve, the eavesdropper, does *intercept-resend*: she catches each pulse with her own receiver in a basis from her sheet, then re-sends what she read with her own laser in that basis (manual §7.5, §8.3). She cannot know Alice's basis in advance. **Predict** on the sifted pulses: half the time Eve's basis equals Alice's, she reads Alice's bit and re-sends it exactly (error 0); half the time it differs, her bit is a coin flip and so is Bob's (error $\frac{1}{2}$):

$$Q = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{1}{2} = 25\%.$$

Measure: the staff insert Eve (manual §7.5). Eve fills her basis row in W3 with a red 6-sided die *before* the run, so what she sees cannot bias her choices (manual §8.3). Repeat Part C steps 1–4 with fresh dice (run 2, W3). No message is encrypted in run 2, so *every* sifted bit is a test bit: $t = S$. **Compare** your QBER with 25 %, and check whether your errors sit only where Eve’s basis differs from Alice’s.

Counterintuitive — and Real

Each test bit catches Eve with probability $\frac{1}{4}$, so she survives t test bits with probability $(3/4)^t$. With $t = 6$ that is **17.8 %**. Pushing it below 1 % takes $t = 17$ ($(3/4)^{17} \approx 0.75$ %). Then 20 key bits + 17 test bits need $S \geq 37$: 52 pulses give that only 0.16 % of the time, while about **85 pulses** give it with 90 % probability.

Why pool the class data. The *standard error* (SE) is the typical size of the random scatter of an estimate around the true value. For the QBER estimate from t test bits, $SE = \sqrt{Q(1-Q)/t}$: at $Q = 0.25$ and $t = 6$, $SE \approx 0.18$, so one group’s estimate can land far from 25 %. Testing every sifted bit shrinks it: at $t = 26$, $SE \approx 0.085$. G groups with t test bits each give $\sqrt{Q(1-Q)/(Gt)}$. Add your counts to the class table and compute the class-pooled QBER: total errors over total test bits.

Real-World Anchor

The **Micius** satellite distributed BB84 keys from space to the ground (2017); **EuroQCI** is the European Commission’s initiative for a quantum communication infrastructure across the EU. Real QKD needs single photons (or *decoy states*: randomly varied pulse intensities that expose a tap), because bright classical light can be tapped.

Instructor demo (Part E, optional)

A non-polarizing beam splitter (NPBS, not in the kit) between Alice and Bob sends part of every pulse to Eve’s receiver and lets the rest through. Bob’s QBER stays 0, yet Eve learns the sifted bits where her basis matched Alice’s — half of them on average, and she knows which. A single photon cannot be split, so this attack needs classical light.

For your report

1. Is the class-pooled QBER of run 2 consistent with 25 % within its standard error? Is your run-1 QBER consistent with 0, and if not, what on the bench could explain it?
2. Why may Alice and Bob announce their bases in public but never their bits? What does Eve learn from the announcements?
3. Part B and the tap demo both rely on classical light splitting. What would change with a true single-photon source?
4. Look only at the first 6 sifted bits of run 2: would a 6-bit test have caught Eve?

Group _____ Names _____ Date _____

	$s_B = 0^\circ$			$s_B = 45^\circ$		
s_A	$\cos^2(s_A + s_B)$	predict	observed	$\cos^2(s_A + s_B)$	predict	observed
-45°						
0°						
45°						
90°						

W2 · Run 1: Alice and Bob, no Eve ($N = 52$ pulses in blocks of $18 + 18 + 16$). Basis: $+$ or $\times$. Bit: 0 or 1 (Bob: transmitted = 0, reflected = 1). Tick matched bases; in the Test row write $=$ or $\neq$ on each test bit.

[illegible][illegible][illegible][illegible]

W3 · Run 2: with Eve (Eve's basis row is filled with her die *before* the run). No encryption in run 2: every matched pulse is a test bit, so write = or $\neq$ on each one.

Pulse	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Eve basis																		
Alice basis																		
Alice bit																		
Bob basis																		
Bob bit																		
Same basis? ✓																		
Test = / $\neq$																		

Pulse	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Eve basis																		
Alice basis																		
Alice bit																		
Bob basis																		
Bob bit																		
Same basis? ✓																		
Test = / $\neq$																		

Pulse	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52
Eve basis																
Alice basis																
Alice bit																
Bob basis																
Bob bit																
Same basis? ✓																
Test = / $\neq$																

W4 · Results. QBER = errors/ t (t = marked Test cells, errors = cells marked $\neq$); SE = $\sqrt{Q(1-Q)}/t$ at $Q = 0.25$; class QBER = total errors / total test bits of all G groups, pooled SE = $\sqrt{Q(1-Q)/(Gt)}$.

	S	S/N	t	errors	QBER	SE ($Q=0.25$)	$(3/4)^t$	class QBER $\pm$ pooled SE
Run 1 (no Eve)								
Run 2 (Eve)								

W5 · Encryption (only if run 1 showed 0 test errors). Ciphertext $y = m \oplus k$; Bob recovers $m = y \oplus k$.

Letter																
Message bits m																
Key bits k																
Ciphertext y																

5-bit alphabet (manual Ch. 9): letter, index, code.

A 0	B 1	C 2	D 3	E 4	F 5	G 6	H 7	I 8	J 9	K 10	L 11	M 12
00000	00001	00010	00011	00100	00101	00110	00111	01000	01001	01010	01011	01100
N 13	O 14	P 15	Q 16	R 17	S 18	T 19	U 20	V 21	W 22	X 23	Y 24	Z 25
01101	01110	01111	10000	10001	10010	10011	10100	10101	10110	10111	11000	11001